

A prototype of this interactive installation is being developed during the "Co-Creating with Art and AI" residency at Mila Institute. It will be shown as a group exhibition on September 14th and 15th at the Agora du Coeur des Sciences in Montreal. The residency explores how we can assert agency in designing spaces and practices where creative processes, AI thinking, and human-centered design coexist. Specifically, it addresses ethical considerations such as transparency, consent, privacy, bias, and the ecological impacts of AI, while also seeking to develop alternative approaches to AI development beyond the traditional Western frameworks.

Artist and collaborators:

Mariane Laporte is a Montreal-based interdisciplinary artist and researcher focused on emerging media. She is currently completing a Master's in Communication with a focus on research-creation at UQAM and is actively involved with the Hexagram research network. Mariane's artistic practice spans multiple disciplines: she is the singer-songwriter for the Québécois experimental-electro-rock band Ellemetue, a filmmaker, and a journalist for Panorama-cinema. Her recent work explores interactive art, performance, feminist studies, artificial intelligence, biotechnologies and digital literature.

Ali Saheb-Pasand is a PhD student at McGill University, affiliated with the Mila - Montreal Institute for Learning Algorithms. Ali's research focuses on computational neuroscience, deep learning, reinforcement learning, and representation learning.

Kenny Lefebvre is a researcher and media artist specializing in experimental media. His work involves video projection, electronic prosthesis design, and multimedia installation. He has played key roles in a range of projects, from video design for theater productions to offering technical consulting services. Additionally, he specializes in web integration, SEO, and e-marketing.

Title: *Immaculate Conception Inc.: A Journey Through the Future of Birth*

Imagine a world where the miracle of life transcends the biological constraints of the human body. *Immaculate Conception Inc.* invites you to experience a gestation - from fertilization to delivery - in just fifteen minutes. A critique of assisted procreation in capitalist societies. This interactive installation combines generative AI and the human senses to simulate a pregnancy outside of the woman's womb. This is not just a glimpse into the future—it is a research-creation about ectogenesis, raising profound questions about the essence of motherhood.

Installation Overview:

Participants enter a digital environment filled with multisensory devices equipped with screens, each capable of conceiving a baby. As they engage with these interactive machines, they speed-connect with their future child at all stages: from fertilization, to embryonic development, and fetal growth, culminating in a computer-generated labor and delivery. Guided by a narrator, their mission is to ensure healthy development, a journey enhanced by integrated audio, visual, and tactile elements. AI-driven imagery is generated in real-time via an API, while speech-to-text technology activates system prompts and language models, responding to participants' inputs. Python coding integrates these interactive components. The journey begins with the signing of an adoption contract and concludes with the issuance of a birth certificate, leaving the users with a provocative question: What if you realized that the connection with the newborn was never truly formed? Would you still choose to adopt?

Scene 1: Introduction

A screen welcomes a user, displaying a fifteen-minute countdown. The multisensorial device introduces "Immaculate Conception Inc.," a fictional medical company specializing in ectogenesis. Once the procedure begins, there is no turning back and skipping a step results in the loss of parental insurance.

Scenes 2 to 9: Sensory Stations

Station 1: The participant uses an iPad to connect an augmented umbilical cord to their navel, experiencing nutrient and oxygen transfer to the fetus,

Station 2: A microphone allows the user to tell a story to the fetus, with a brain barometer filling up, symbolizing memory formation.

Station 3: The user peers through a small opening to uncover a dream world, where the fetus' subconscious is generated based on prompts from station 2—a fusion of the user's story with the one of past user. The randomness of the dream is influenced by the user's body temperature.

Station 4: A video-game-like gender reveal involves popping a balloon to discover the fetus gender, represented by confetti symbolizing a diversity of gender symbols. The user chooses a name.

Station 5: The user pursues a song, according to the past user's melody, emphasizing the importance of frequencies for fetal development. The voice is recorded with a microphone and modified via a MIDI synth.

Station 6: A camera adjusts the clarity of the image it captures of the user to simulate the developing vision.

Station 7: The user feels a tactile connection with the fetus' heart through a soft fabric, reinforcing the emotional bond.

Station 8: A labor simulation allows the user to "push" and reveal the baby's face, composed of a morph of features from previous users and the current participant (the user's portrait was captured at station 6).

Station 9: The user - now parent - tries to calm the cries of the little one while having to keep an eye on a baby monitor equipped with night vision mode, while waiting to be called to reception. If needed, Stations 5 and 9 are available to help soothe the baby.

Scene 10: Conclusion

The new parent chooses to either adopt, abandon or return the baby to the cloud.

Note: All user inputs are compiled into a local database, creating a self-sustaining ecosystem where future interactions are influenced solely by past users' data—eliminating the need for large-scale data collection and ensuring maximum privacy. This strategy will allow all users to co-create a baby within the ever-evolving installation.

1. Ectogenesis and the Development of Artificial Embryos

The installation proposes a simulation of what could be ectogenesis; a human gestation that takes place entirely outside the woman's body. While artificial wombs are not yet used for human reproduction, significant progress has been made in this field, particularly with research on artificial embryos and incubation systems for extreme preemies.

2. Fetal Consciousness and Sensory Stimuli

Another fundamental aspect of the installation is the focus on sensory stimuli during gestation. Research shows that fetal brain development is already advanced halfway through gestation and that sensory stimuli play a crucial role in correctly wiring primary sensory regions. The installation allows participants to interact with a simulated fetus using tactile, auditory, and visual sensors, thus replicating some of the stimuli the fetus might experience in utero. This approach questions how technology can not only replicate but also potentially alter these crucial sensory experiences for best or worse.

3. Reproductive Ethics and Biotechnology

Finally, the project fits into the broader debate on the ethical implications of new reproductive technologies, the significance of parenthood, biological inheritance, and the very definition of humanity. The interactive installation allows a space where users can not only tame this perhaps not so far reality but also actively co-create a simulated life.